OPENEGIZ • TEAM 075

EcoMine

Intelligent Green Mining

Digital Twin

Live truck telemetry / environmental risk intelligence / digital twin synchronization

Competition presentation • October 2026

The challenge

Environmental visibility across mining haulage operations

Fragmented signals

Emissions, energy, water and production data can be difficult to interpret together.

Delayed awareness

High-emission conditions need to be identified clearly as telemetry arrives.

Disconnected systems

Monitoring and digital-twin representations often require integration.

Our solution

A connected, transparent monitoring prototype

Stream

Simulate four mining trucks and publish telemetry continuously through MQTT.

Understand

Store measurements, classify risk, synchronize a Ditto Thing and visualize trends.

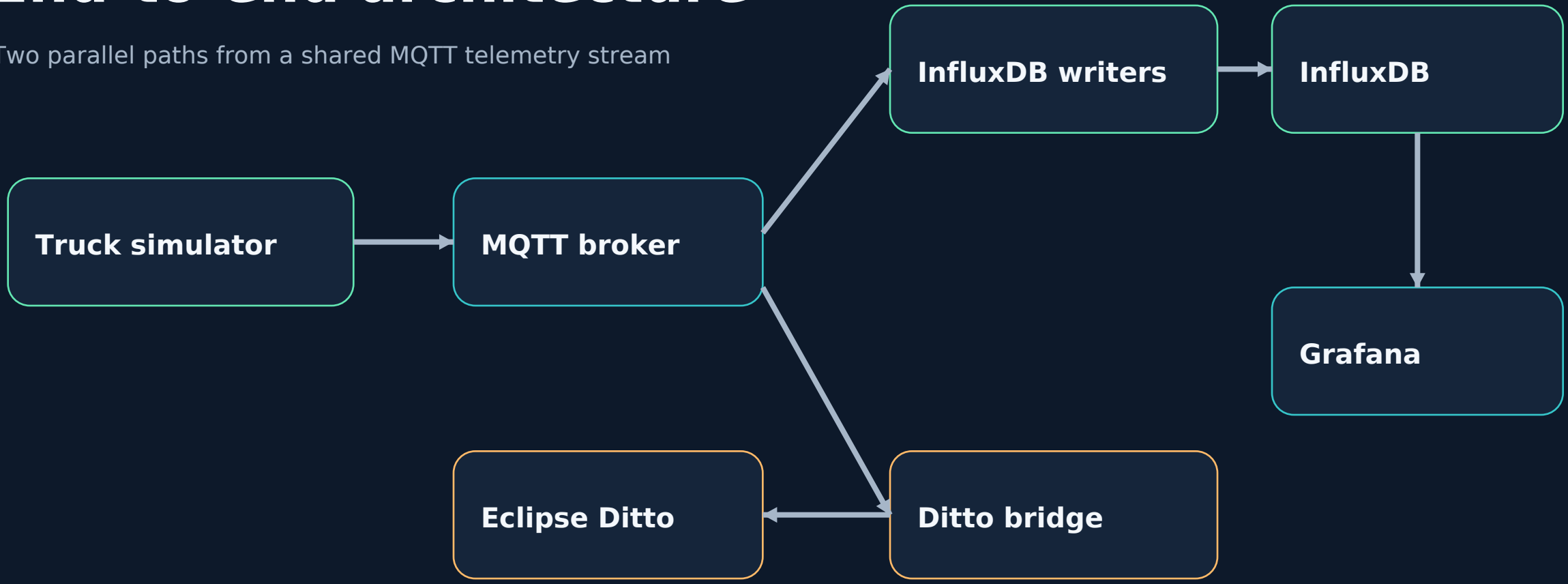
4 simulated trucks

5 Grafana panels

Live digital twin

End-to-end architecture

Two parallel paths from a shared MQTT telemetry stream



Python • Docker Compose • Mosquitto • InfluxDB • Eclipse Ditto • Grafana

Telemetry model

Each event describes one simulated truck reading

truck_id	Truck identity
truck_type	Equipment category
ore_tonnes	Ore transported
energy_kwh_equivalent	Energy equivalent
water_m3	Water use
co2_kg	CO2 emissions
timestamp	Event time

Environmental risk engine

Illustrative thresholds for synthetic CO2 readings

NORMAL

Below 150 kg

WARNING

150 to <200 kg

CRITICAL

200 kg and above

Risk values are saved to InfluxDB and can trigger a Grafana alert. These limits are demonstration rules, not regulatory standards.

Live monitoring

Five panels configured in Grafana

CO2 emissions

kg



Water consumption

m³



Energy consumption

kWh equivalent



Ore transported

tonn



Environmental risk

level



Grafana dashboard is exported as JSON and provisioned from the repository.

Digital twin synchronization

A live representation through Eclipse Ditto

Digital twin ID

ecomine:smart-mine

The bridge updates the Thing features through the Ditto HTTP API.

Verified behavior

Bridge logs report DITTO UPDATED with HTTP 204 responses for truck telemetry.

One shared Thing with aggregated/latest telemetry; not a complete physical mine model.

Verified demonstration

What was observed in the local OpenEgiz deployment

✓ **Docker services**	Four EcoMine containers running
✓ **InfluxDB**	Telemetry and risk writes return HTTP 204
✓ **Eclipse Ditto**	Digital twin updates return HTTP 204
✓ **Grafana**	Five panels visible; dashboard auto-loaded
✓ **Alerting**	Critical risk alert observed FIRING

Deployment and reproducibility

Repository-backed Docker services and Grafana dashboard

Deployment

EcoMine has a separate Docker Compose overlay and a Python container image.

Configuration

Grafana dashboard JSON and provisioning YAML are included in the repository.

Clean-machine deployment and alert auto-provisioning remain to be validated. Credentials must be supplied securely.

Scope and roadmap

Clear boundaries, practical next steps

Current scope

Synthetic truck telemetry
Rule-based CO2 risk
One aggregated Ditto Thing
Local monitoring and alerting

Next milestones

Reproducible clean install
Automatic alert provisioning
Interactive what-if scenarios
Validation with real equipment data

EcoMine

See the mine. Understand the impact.

A working prototype for connected environmental intelligence in mining operations.

SOURCE CODE & DOCUMENTATION

github.com/openegiz-contest/team-075

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